

Information Theory and Cognition

Problem Set 1: Probability theory warm-up

Due: Friday 2/6 (before midnight)

1. Let (Ω, P) be a probability space, and A, B, C are events. Prove or disprove:
 - (a) If A and B are independent, then A and $\bar{B}$ are also independent.
 - (b) If A, B and C are pairwise independent, then $P(A \cap B \cap C) = P(A)P(B)P(C)$.
 - (c) If A and B are independent then they are also conditionally independent, i.e., $P(A|B, C) = P(A|C)$.
2. There are three biased coins with biases $\theta_1 = 0.25$, $\theta_2 = 0.5$ and $\theta_3 = 0.75$. A coin is picked at random and is tossed three times.
 - (a) What is the probability that heads will come up in all three tosses?
 - (b) Suppose that you tossed the coin and heads came up in all three tosses. What is the probability that the coin corresponds to θ_2 ?
 - (c) Suppose that heads came up in all three tosses. What is the probability that heads will come up if you toss the coin once again?
3. Show that there does not exist any number c such that the following function would be a probability function:

$$f(x) = \begin{cases} \frac{c}{x} & x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$
4. Suppose a crime has been committed. Blood is found at the scene for which there is no innocent explanation. It is of a type which is present in 1% of the population.
 - (a) The prosecutor claims: “There is a 1% chance that the defendant would have the crime blood type if he were innocent. Thus there is a 99% chance that he is guilty.” This is known as the *prosecutor’s fallacy*. What is wrong with this argument?
 - (b) The defender claims: “The crime occurred in a city of 800,000 people. The blood type would be found in approximately 8000 people. The evidence has provided a probability of just 1 in 8000 that the defendant is guilty, and thus has no relevance.” This is known as the *defender’s fallacy*. What is wrong with this argument?